
UNIT 18 FISCAL POLICY*

Structure

- 18.0 Objectives
- 18.1 Introduction
- 18.2 Goals of Fiscal Policy
- 18.3 Discretionary Fiscal Policy
 - 18.3.1 Discretionary Fiscal Policy to Cure Recession
 - 18.3.2 Discretionary Fiscal Policy to Control Inflation
 - 18.3.3 Time Lags in Implementation of Discretionary Fiscal Policy
- 18.4 Non-Discretionary Fiscal Policy
 - 18.4.1 Income Taxes
 - 18.4.2 Transfer Payments
 - 18.4.3 Discretionary vs. Non-Discretionary Measures
- 18.5 Let us Sum Up
- 18.6 Answers to Check Your Progress Exercises

18.0 OBJECTIVES

After going through the unit you will be in a position to

- explain the concepts of fiscal policy, government budget, and budget balance;
- distinguish between Discretionary and Non-Discretionary fiscal policies;
- identify the role of these policies in reducing output fluctuations; and
- explain the issues related to the implementation of fiscal policy.

18.1 INTRODUCTION

You may be aware by now that the economy does not always work smoothly. There often occur fluctuations in the level of economic activity. At times the economy finds itself in the grip of recession when levels of income, output and employment are below their full employment levels. At other times the economy is overheated, which means occurrence of rising prices and high inflation. The classical economists believed that an automatic mechanism works to restore stability in the economy; recession would cure itself and inflation will be automatically controlled. However, the empirical evidence during the 1930s when severe depression took place in the western capitalist economies shows that no such automatic mechanism works to bring about stability in the economy. In view of this Keynes argued for intervention by the government to cure depression and inflation by adopting appropriate tools for macro-economic policy. The two important tools of macroeconomic policy are fiscal policy and monetary policy. We have already dealt with monetary policy in the previous two units.

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In the present unit we will dwell upon the role of fiscal policy for stabilizing the economy.

18.2 GOALS OF FISCAL POLICY

Fiscal policy basically is demand management policy in the sense that it influences aggregate demand through changes in government expenditure. Taxes, transfer payments, and government expenditures on goods and services are the tools of fiscal policy. The main objective of fiscal policy is to stabilize output by managing aggregate demand, keeping output close to potential output, and reducing the size and duration of business cycle fluctuations. This requires changes in the government's expenditure plans and tax policy to offset changes in autonomous consumption, investment and exports that would otherwise push the economy away from equilibrium at potential output.

Fiscal Policy tools, taxes, transfers and government expenditure, influence aggregate demand through disposable income and consumption expenditure, and directly through government expenditures. The net tax rate also affects the size of the expenditure multiplier (also called Keynesian multiplier or spending multiplier). Thus, changes in tax rates, transfer rates, and government expenditures shift the AD curve.

At the time of recession the Government increases its expenditure or cuts down taxes or adopts a combination of both. On the other hand, to control inflation the Government cuts down its expenditure or raises taxes. In other words, to cure recession, expansionary fiscal policy and to control inflation, contractionary fiscal policy is adopted. Adoption of such expansionary fiscal policy will cause deficit in the government's budget and contractionary fiscal policy will cause a surplus in the government's budget. As the government implements fiscal policy through its budget, let us first understand the budget balance.

The government budget describes what goods and services the government will buy during the coming year, what transfer payments it will make, and how it will pay for them. When revenues and spending are equal, the budget is balanced. When revenues exceed spending, there is a budget surplus. When revenues fall short of spending, there is a budget deficit. The government budget balance is the difference between revenues and expenditures.

$$\text{Government Budget Balance (BB)} = NT - G = (tY - G)$$

Where G = Government Purchases, NT = net Tax Revenue and, t = tax rate

The budget balance, whether deficit, surplus or zero, is determined by three things:

1. The *net tax rate* t set by the government;
2. The level of *expenditure* G set by the government; and
3. The level of *output* Y determined by AE and AD

In Fig. 18.1, panels (a) and (b), we present the budget function and the budget balance respectively. In Fig 18.1 panel (a), the budget is balanced (i.e., $G=NT$) at the aggregate income level Y_1 . The budget has a deficit for income level less than

Y_1 (i.e., $Y < Y_1$) and a surplus for income level more than Y_1 (i.e., $Y > Y_1$). In Fig. 18.1 panel (b), the budget balance (BB) is determined by the level of GDP. A budget function shows us the different budget balances for one fiscal policy program at different levels of national income. Finally, notice that because a budget function describes one fiscal plan, any change in the fiscal plan will change the BB line to show a new budget function. We observe that $BB=0$ (i.e., $NT=G$), for income level, Y_1 . Correspondingly, $BB>0$ for income levels greater than Y_1 and $BB<0$ for income levels less than, Y_1 .

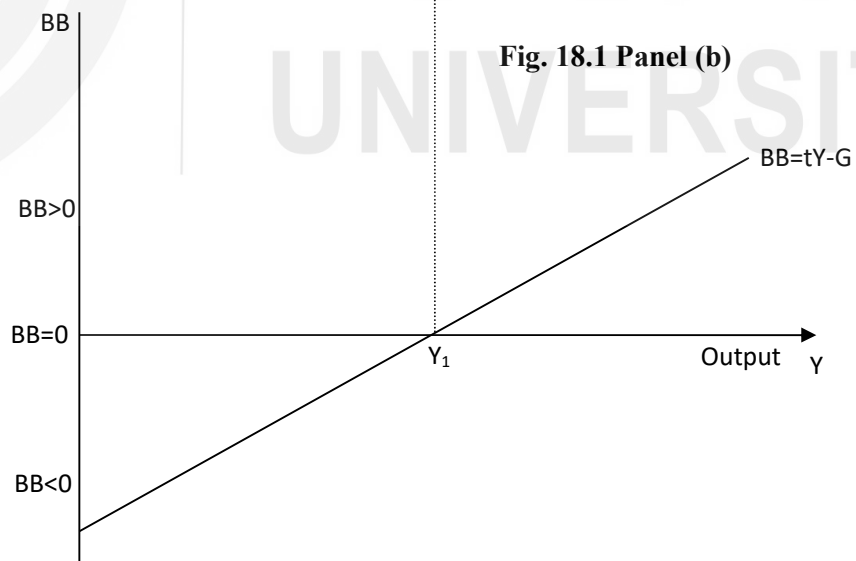
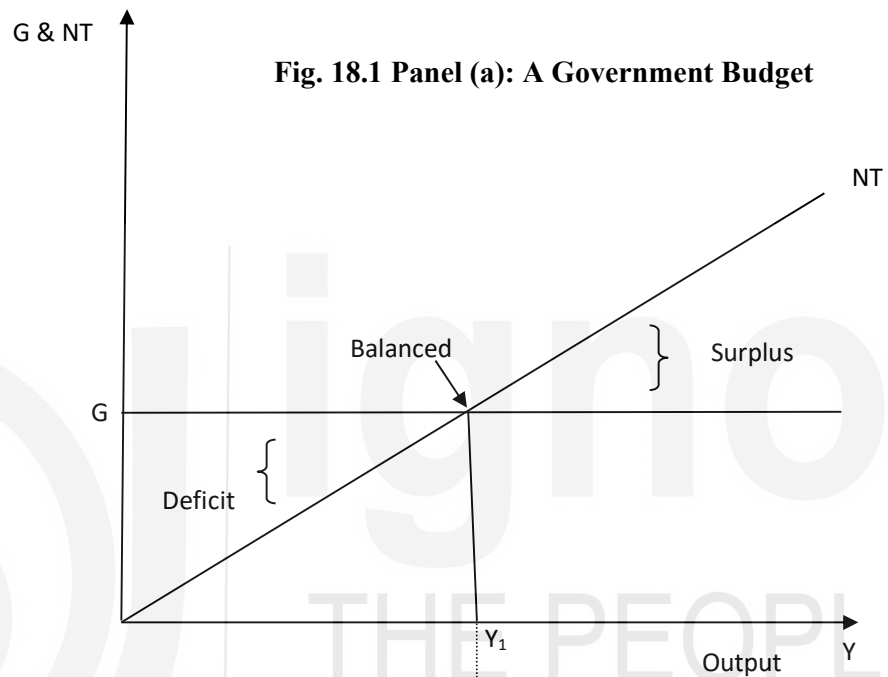


Fig. 18.1 shows the government budget is in balance at income level, Y_1 . It is in surplus to the right of Y_1 ($Y > Y_1$,) and in deficit to the left of Y_1 ($Y < Y_1$,).

- 1) What is meant by fiscal policy?

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- 2) Explain the type of fiscal policies governments use to control recession and inflation?

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18.3 DISCRETIONARY FISCAL POLICY

Fiscal policy is of two kinds, viz., i) Discretionary fiscal policy, and ii) Non-Discretionary fiscal policy of automatic stabilizers. By discretionary fiscal policy we mean deliberate change in government expenditure and taxes to influence the level of national output and prices. These are non-mandatory changes in taxation, spending, or other fiscal activities by a government in response to economic events or changes in economic conditions. Discretionary fiscal policy implies government actions above and beyond existing fiscal policies, and often occurs in periods of recession or economic turbulence. Discretionary fiscal policy relates to demand management that uses government spending and taxation policy to influence aggregate demand. The government uses an expansionary fiscal policy to correct a recessionary gap and a contractionary fiscal policy to correct an inflationary gap.

18.3.1 Discretionary Fiscal Policy to cure Recession

The recession in an economy occurs when aggregate demand decreases below the full employment level due to a fall in investment, consumption or net exports. This fall in aggregate demand below the full employment level creates a deflationary or recessionary gap. It is the task of fiscal policy to close this gap by increasing government expenditure or reducing the taxes. We discuss below policy instruments used to cure recession.

a) Increase in Government Expenditure

For a discretionary fiscal policy to cure depression, the increase in government expenditure is an important tool. Government may increase expenditure by starting public works. For undertaking all these public works, government buys various types of goods and materials and employs workers. The effect of this increase in expenditure is both direct and indirect. The direct effect is the increase in incomes of those who sell materials and supply labour for these projects. The output of these public works also goes up together with the increase in incomes. Not only that, Keynes showed that increase in government expenditure also has an indirect effect in the form of the working of a multiplier (see Block 1, Unit 2). Those who get more income spend them further on consumer goods depending on their marginal propensity to consume. The impact of increase in government expenditure in a recessionary condition is illustrated in Fig 18.2. The aggregate expenditure is given by the curve AE while the 45° line depicts the equality between aggregate output (Y) and AE. Suppose the initial aggregate expenditure, AE_0 , intersect the 45° line at point E_1 and thereby determine Y_0 as the level of output. As this level of equilibrium output is below the full employment level (Y_f), a deflationary gap of Y_0Y_f is created. We observe from Fig. 18.2 that, the deflationary gap can be eliminated if the government increases its expenditure by E_1H . The aggregate expenditure curve will shift upwards to AE_1 and as a result the equilibrium level of income will increase to the full employment level, Y_f . Note that the increase in output by Y_0Y_f is more than the increase in government expenditure $\Delta G = E_1H$, due to multiplier effect.

Fig. 18.2: Increase in Government Expenditure

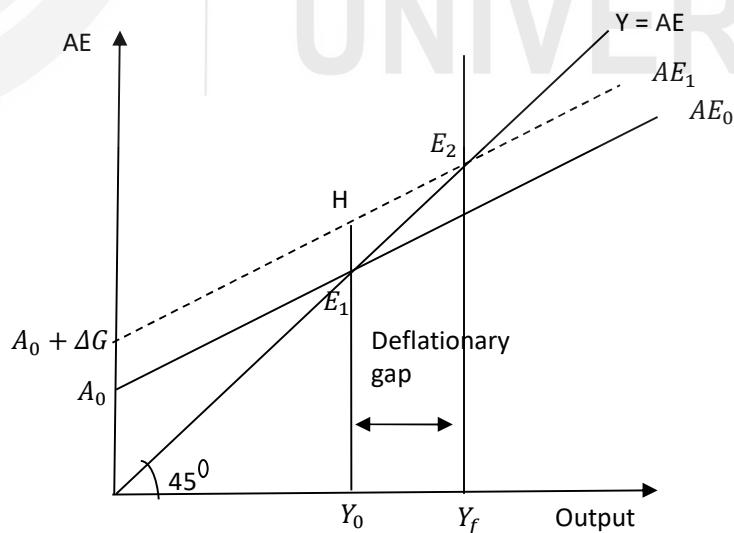


Fig.18.2 shows that an increase in Government expenditure increases Y and closes the deflationary gap.

b) Reduction in Taxes

An alternative fiscal measure to overcome recession and to achieve expansion in output and employment is reduction in taxes. The reduction in taxes increases the disposable income of the society and causes the increase in the consumption spending of the people. Thus reduction in taxes will cause an upward shift in the consumption function. If along with the reduction in taxes, the government expenditure is kept unchanged, aggregate expenditure curve will shift upward due to rise in consumption function curve. This will have an expansionary effect and the economy will be lifted out of recession, and national income and employment will increase. Fig. 18.3 shows the effect of a reduction in taxes on addressing the problem of recessionary gap. Aggregate demand curve AE_0 , intersect the 45° line at point E_1 , with Y_0 , as the equilibrium level of income. A recessionary gap of $Y_0 Y_f$ exists as this level of output falls short of the full employment level, Y_f . A fall in the tax rate, increases the slope of the aggregate expenditure curve (by increasing the value of the multiplier) and shifts the aggregate expenditure curve outward to AE_1 , which intersects with the $Y=AE$ line at point E_2 . As the level of equilibrium income now becomes equal to the full employment level of income, Y_f , the recessionary gap disappears. Thus, a reduction in taxes will increase consumption expenditure C , and shifts the aggregate expenditure curve upward and will result in, through the working of the multiplier, a higher level of equilibrium level of income.

Fig 18.3: Increase in Net Tax Rate

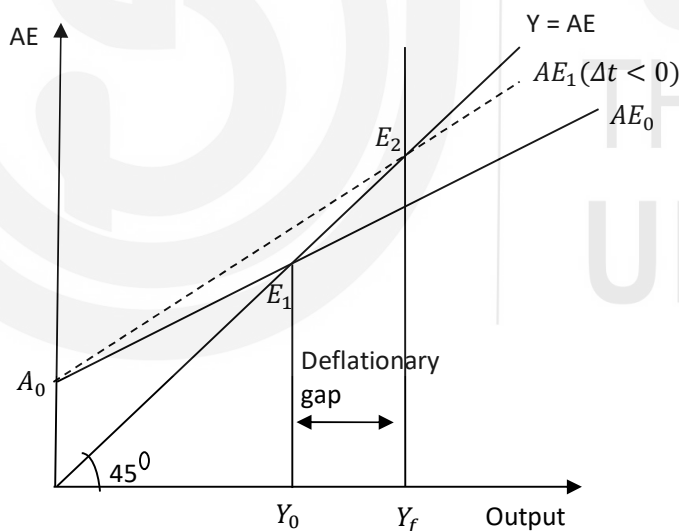


Fig 18.3 shows that a decrease in net tax rate increases spending and the multiplier to close the recessionary gap.

18.3.2 Discretionary Fiscal Policy to Control Inflation

A situation of excess demand in the economy can arise because of three reasons: i) increase in household demand due to large increase in consumption, ii) increase in demand by firms due to increase in investment, and iii) increase in government expenditure. As aggregate expenditure increases beyond what the

economy can potentially produce by fully employing its given resources, the situation of excess demand builds inflationary pressures in the economy. An inflationary situation can also arise if too large an increase in money supply in the economy occurs. In these circumstances, an inflationary gap occurs which tend to bring about rise in prices. Under such circumstances anti-cyclical fiscal policy calls for reduction in aggregate expenditure. Thus fiscal policy measures to control inflation are (a) reducing government expenditure (b) increasing taxes.

a) Reducing Government Expenditure

Fig. 18.4 shows how the reduction in government expenditure will help checking inflation. You can see from the figure that an aggregate demand, AE_0 intersects the 45° line at point, E_1 , and determines equilibrium level of income at $Y_0 > Y_f$, (the full employment level). As a result, an inflationary gap equal to $Y_0 - Y_f$ emerges. A reduction in government expenditure equal to E_1H , reduces aggregate expenditure and shifts the curve down to AE_1 , which will restore the equilibrium at the full employment level, Y_f . The reduction in Government expenditure equal to E_1H through the operation of multiplier will result in a multiple decline in the level of national income or output. Fig. 18.4 shows that the decrease in government expenditure by E_1H has led to a much bigger decline in output by $Y_0 - Y_f$.

Fig. 18.4: Decrease in Government Expenditure

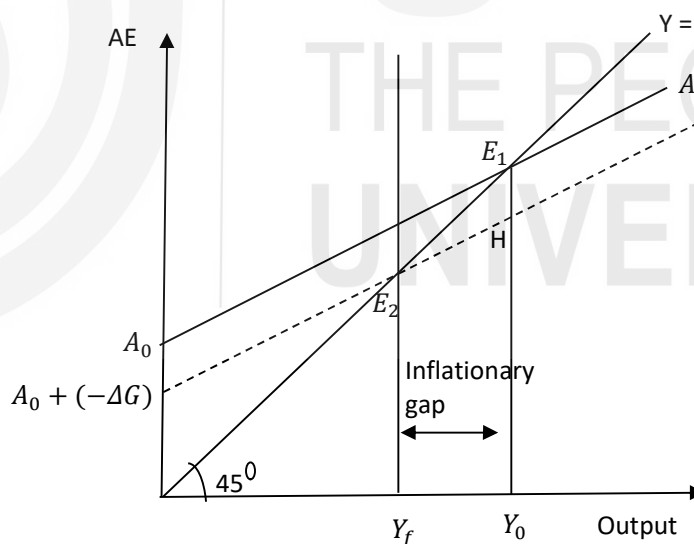


Fig. 18.4 shows that a decrease in Government expenditure decreases Y and closes the inflationary gap.

b) Increase in Taxes

In addition to the reduction in Government expenditure, the government can increase taxes to reduce aggregate demand. The hike in taxes reduces the disposable income of the people and thereby forces them to reduce their

consumption demand. In Fig. 18.5, the inflationary gap of $Y_0 Y_f$ is closed by an increase in the net tax rate which reduces the slope of AE and shifts the aggregate expenditure downward to AE_1 . This downward shift of the aggregate expenditure curve moves the economy back to the full employment equilibrium at, Y_f .

Fig. 18.5: Increase in Net Tax Rate

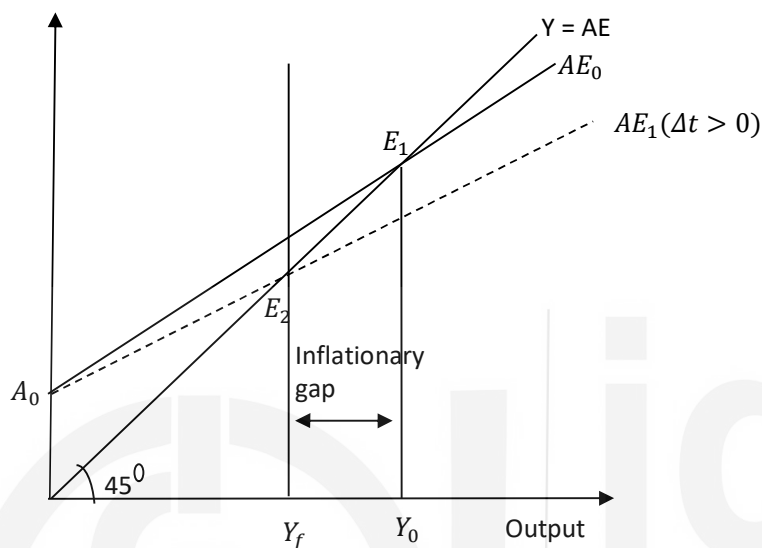


Fig. 18.5 shows that an increase in net tax rate reduces spending and the multiplier to close the inflationary gap.

18.3.3 Time Lags in Implementation of the Discretionary Fiscal Policy

Although the government has long conducted monetary and fiscal policy, the view that it should use these policy instruments to try to stabilize the economy is more recent. To many economists the case for active government policy is clear and simple. Recessions are periods of high unemployment, low incomes, and increased economic hardship. The model of aggregate demand and aggregate supply shows how shocks to the economy can cause recessions. It also shows how monetary and fiscal policy can prevent (or at least soften) recessions by responding to these shocks. These economists consider it wasteful not to use these policy instruments to stabilize the economy. Other economists are critical of the government's attempts to stabilize the economy. These critics argue that the government should take a hands-off approach to macroeconomic policy. Why do these critics want the government to refrain from using monetary and fiscal policy for economic stabilization? To find out, let's consider some of their arguments.

Stabilization goals are not easy to achieve. The existence of various kinds of time lags, or delays in the response of the economy to stabilization policies, can make the economy difficult to control. Indeed, the problem for policymakers is even more difficult, because the lengths of the lags are hard to predict. Economists distinguish between two lags that are relevant for the conduct of stabilization policy: the inside lag and the outside lag. The inside lag is the time between a shock to the economy and the policy action responding to that shock. This lag arises because it takes time for policymakers first to recognize that a shock has occurred and then to put appropriate policies into effect. The outside lag or the response lag is the time between a policy action and its influence on the economy. This lag arises because policies do not immediately influence spending, income, and employment. The inside lag can be further classified into the recognition lags and the implementation lags.

a) Recognition Lag

It takes time for policy makers to recognize a boom or a slump. Many important data—those from the national income and product accounts, for example—are available only quarterly. It usually takes several weeks to compile and prepare even the preliminary estimates for these figures. If the economy goes into a slump on January 1, the recession may not be detected until the data for the first quarter are available at the end of April.

b) Implementation Lag

The problems that lags pose for stabilization policy do not end once economists and policy makers recognize that the economy is in a boom or a slump. Even if everyone knows that the economy needs to be stimulated or reined in, it takes time to put the desired policy into effect, especially for actions that involve fiscal policy. Implementation lags result.

c) Response Lag

Even after a macroeconomic problem has been recognized and the appropriate policies to correct it have been implemented, there are response lags – lags that occur because of the operation of the economy itself. For example, it takes time for the multiplier to reach its full value. The delay in the multiplier process occurs because neither individuals nor firms revise their spending plans instantaneously. Until they can make those revisions, extra government spending does not stimulate extra private spending.

In practice, it takes about a year for a change in taxes or in government spending to have its full effect on the economy. This response lag means that if we

increase spending to counteract a recession today, the full effects will not be felt for 12 months. By that time, the state of the economy might be very different.

The long and variable lags associated with fiscal policy certainly make stabilizing the economy more difficult. Advocates of passive policy argue that, because of these lags, successful stabilization policy is almost impossible. Indeed, attempts to stabilize the economy can be destabilizing. Suppose that the economy's condition changes between the beginning of a policy action and its impact on the economy. In this case, active policy may end up stimulating the economy when it is heating up or depressing the economy when it is cooling off. Advocates of active policy admit that such lags do require policymakers to be cautious. But, they argue, these lags do not necessarily mean that policy should be completely passive, especially in the face of a severe and protracted economic downturn.

Check Your Progress 2

- 1) Show diagrammatically how a recession can be controlled through the use of discretionary fiscal policy.

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18.4 NON-DISCRETIONARY FISCAL POLICY

Non-discretionary fiscal policy of automatic stabilizers is a built-in tax or expenditure mechanism that automatically increases aggregate demand when recession occurs and reduces aggregate demand when there is inflation in the economy without any special deliberate actions on the part of the government.

These policies, called automatic stabilizers, are designed to reduce the lags associated with stabilization policy. Automatic stabilizers are policies that stimulate or depress the economy when necessary without any deliberate policy change. In the non-discretionary fiscal policy, the tax structure and expenditure pattern are so designed that taxes and government spending vary automatically in appropriate direction with the changes in national income. That is, these taxes and expenditure pattern without any special deliberate action by the government

and parliament, automatically raise aggregate expenditure in times of recession and reduce it in times of boom, thus ensuring economic stability. These fiscal measures are therefore called automatic stabilizers or built-in stabilizers. Since these stabilizers are built into the budget and do not require any deliberate action by the government, they represent non-discretionary fiscal policy. Built in stability of tax revenue and government expenditure of transfer payments and subsidies is created because they vary with national income. These taxes and transfers automatically bring about appropriate changes in the aggregate expenditure and reduce the impact of recession and inflation. This means that recession and inflation will be less intense with the operation of the automatic stabilizers. Let us understand how taxes and transfers work to reduce the fluctuations of business cycles.

18.4.1 Income Taxes

At given net tax rates, a fall in national income, output, and employment reduces tax collections. In other words, when national income declines at times of recession, the tax revenue declines as well. This prevents aggregate expenditure from falling by the same proportion as the decline in income because the disposable income changes by less than the change in national income. The slope of the aggregate expenditure function $(c(1-t)-m)$ is lower, and so is the multiplier. Conversely, in a boom, net tax revenues rise and disposable income rises by less than the rise in national income, which helps dampen the boom.

18.4.2 Transfer Payments

When there is recession and as a result unemployment increases, the Government has to spend more on compensation for unemployment and other welfare programmes such as food stamps, rent-subsidies, subsidies to farmers. This hike in Government expenditure tends to make recession short-lived and less intense. On the other hand, when at times of boom and inflation, national income increases and therefore unemployment falls, the Government curtails its programme of social benefits which results in lowering Government expenditure. This smaller government spending helps to rein in inflation.

18.4.3 Discretionary vs. Non-Discretionary Measures

It follows from the above discussion that automatic stabilizers reduce the intensity of business fluctuations that is, both recession and inflation. By reducing the sensitivity of the economy to expenditure shocks, automatic stabilizers are always at work reducing the size of output and employment fluctuations. However, automatic stabilizers only serve to moderate the fluctuations in real GDP caused by fluctuations in autonomous expenditure. They do not offset those autonomous expenditure disturbances. There is no automatic

change in autonomous government expenditure or tax rates. Those changes usually come from discretionary fiscal policy.

Fig 18.6 shows how the budget balance gets affected as a result of the operation of non-discretionary and discretionary fiscal policies. A government budget function $BB_0 = t_0Y + G_0$ and a structural budget balance SBB_0 at potential output (Y_f) is shown in Fig. 18.6. This budget function represents a fiscal program. Automatic stabilization is a part of this program. It comes from the *slope of the budget function*, the net tax rate t_0 in this case. Automatic stabilization works by changing the budget balance along the BB_0 line as income fluctuates between Y_1 and Y_2 . On the other hand, discretionary stabilization shifts the budget function as a result of changes in government expenditure or taxes.

Fig 18.6: Automatic and Discretionary Fiscal Policy

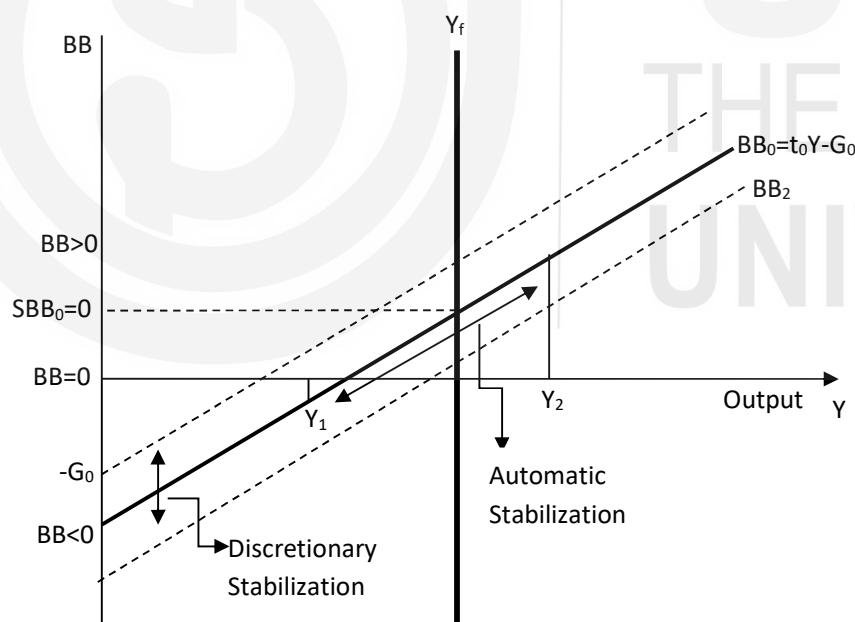


Fig 18.6 shows *Automatic stabilization* comes from changes in the budget balance along the BB_0 line as Y fluctuates between Y_1 and Y_2 . *Discretionary stabilization* shifts the budget function as a result of changes in government expenditure or taxes.

Check Your Progress 3

- 1) What is the difference between discretionary and non-discretionary fiscal policy?

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- 2) What problems does the government face in implementation of discretionary fiscal policies?

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18.5 LET US SUM UP

By fiscal policy we mean automatic and discretionary changes by the government in taxation, public expenditure and borrowing in order to achieve the intended goals of economic growth, full employment, income equality and the stabilization of the economy in its growth path. To carry out these objectives government make use of discretionary and non-discretionary policy measures. Non discretionary policy measures operate automatically and get triggered by the state of the economy. On the other hand, discretionary fiscal policy requires deliberate action by the government to manage aggregate demand. In this chapter we have examined whether policy should take an active or passive role in responding to economic fluctuations. There are many arguments on both sides of these questions. Perhaps the only clear conclusion is that there is no simple and compelling case for any particular view of macroeconomic policy. The existence of various kinds of time lags, or delays in the response of the economy to discretionary policies, can make the economy difficult to control. Automatic stabilizers on the other hand do not suffer from time lags as they are built into the budget program by setting the net tax rate, and work automatically. But automatic fiscal stabilizers only reduce the impact of changes in autonomous expenditure. They do not offset them. Hence, both policy measures should be used in tandem to reduce the fluctuations in output and prices.

18.8 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Fiscal policy is demand management policy. Taxes, transfer payments, and government expenditures on goods and services are the tools of fiscal policy.
- 2) At the time of recession the Government increases its expenditure or cuts down taxes or adopts a combination of both. On the other hand, to control inflation the Government cuts down its expenditure or raises taxes.

Check Your Progress 2

- 1) Refer to Fig. 18.2 and Fig. 18.3

Check Your Progress 3

- 1) By discretionary fiscal policy we mean deliberate change in government expenditure and taxes to influence the level of national output and prices. On the other hand, non discretionary fiscal policy of automatic stabilizers is a built-in tax or expenditure mechanism that automatically increases or decreases the aggregate demand as desired.
- 2) The existence of various kinds of time lags such as recognition lags, implementation lags, and response lags create problems in successful implementation of discretionary fiscal policies.

UNIT 19 FISCAL SUSTAINABILITY*

Structure

- 19.0 Objectives
- 19.1 Introduction
- 19.2 Government's Budget Constraint and Budget Deficit
- 19.3 Current versus Future Taxes
 - 19.3.1 Full Repayment in Year 2
 - 19.3.2 Full Repayment in Year t
 - 19.3.3 Debt Stabilization in Year t
 - 19.3.4 Debt-GDP Ratio
- 19.4 The Dangers of High Debt
 - 19.4.1 Debt Default
 - 19.4.2 Lessons from the Past
- 19.5 Money Finance
- 19.6 Effects of a Decrease in Budget Deficit
- 19.7 Let Us Sum Up
- 19.8 Answers/ Hints to Check Your Progress Exercises

19.0 OBJECTIVES

After going through this unit you will be in a position to

- derive the fiscal sustainability condition from the government budget constraint;
- find out the effects of a fall in taxes and a rise in debt ratio on fiscal sustainability;
- explain the consequences of high debt, debt default and debt monetization; and
- identify the short run and long run effects of a decrease in budget deficit.

19.1 INTRODUCTION

Fiscal sustainability is defined as the government's ability to meet its current and future obligations, to maintain its current policies and satisfy its lifetime budget constraint without defaulting on its debt obligations. It is thus the ability of a government to maintain public finances at a credible and serviceable position over the long term. Ensuring long-term fiscal sustainability requires that governments engage in continual strategic forecasting of future revenues and liabilities, environmental factors and socio-economic trends in order to adapt

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financial planning accordingly. High and increasing debt levels are harmful to governments' fiscal positions and can cause a vicious cycle of growing debt, reducing the potential for economic growth as funds are diverted away from productive investments. One concept of fiscal sustainability relates to a government's ability to indefinitely maintain the same set of policies while remaining solvent. If a particular combination of fiscal and monetary policies would, if indefinitely maintained, lead to insolvency, these policies would be referred to as unsustainable. In case the government cannot service the debt (means repay the principal amount and interest accrued), there is a default on the part of the government. Governments often change their policies if it becomes clear that the government is heading towards a default on its debt. Thus, the focus of fiscal sustainability analysis is not on default in debt servicing, which governments tend to avoid but rather on the policy changes needed to avoid eventual default.

19.2 GOVERNMENT BUDGET CONSTRAINT AND BUDGET DEFICIT

The fundamental building block of fiscal sustainability analysis is the government budget constraint, which is an identity. We must begin with a definition of the budget deficit. One can assess whether fiscal policy is sustainable by examining the size of long-term fiscal imbalances. Doing so provides an early warning of explicit or implicit default and also indicates the magnitude of the adjustments needed to preclude default. The main indicator of a country's fiscal imbalance is its annual official deficit – the difference between annual expenditures (government purchases and transfer payments) and tax and nontax receipts.

We can write the budget deficit in year t as

$$\text{Budget Deficit}_t = rB_{t-1} + G_t - T_t \quad \dots(19.1)$$

All variables are in real terms:

B_{t-1} is government debt at the end of year $t - 1$, or, equivalently, at the beginning of year t ; r is the real interest rate, which we shall assume to be constant here.

Thus,

rB_{t-1} equals the real interest payments on the government debt in year t .

G_t is government spending on goods and services during year t .

T_t is taxes minus transfers during year t .

We can rewrite (19.1) as

$$B_t - B_{t-1} = rB_{t-1} + G_t - T_t \quad \dots(19.2)$$

$$B_t - B_{t-1} = rB_{t-1} - X_t \quad \dots(19.2a)$$

where B_t is the quantity of public debt at the end of period t

$B_t - B_{t-1}$ is the issuance of debt in period t or deficit in period t . If the government runs a deficit, government debt increases. If the government runs a surplus, government debt decreases.

X_t is the primary balance which is equal to $T_t - G_t$

rB_{t-1} is the interest payment on debt where r is the real interest rate

The government budget constraint links the change in government debt to the initial level of debt (which affects interest payments) and to current government spending and taxes. It is often convenient to decompose the deficit into the sum of two terms:

- (i) Interest payment on the debt rB_{t-1}
- (ii) The difference between taxes and Spending ($T_t - G_t$). This term is called the 'primary surplus'.

We can rewrite (19.2a) as

$$B_t = (B_{t-1} + rB_{t-1}) - X_t \quad \dots (19.2b)$$

Intuitively the debt at the end of period t , that is B_t , is equal to the sum of debt in period $(t - 1)$, that is B_{t-1} , and the interest payments on this debt in period $(t - 1)$, that is rB_{t-1} , less the primary surplus in period t , that is X_t . The net debt in period t , $(B_t - B_{t-1})$ is thus equal to the interest payments on the debt in period $(t - 1)$, that is B_{t-1} , less the primary surplus in period t , that is X_t .

Check Your Progress 1

- 1) Explain the concept of government's budget constraint.

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19.3 CURRENT VERSUS FUTURE TAXES

The sources of funds for a government are limited. There are two sources of revenue for the government, viz., tax revenue and non-tax revenue, of which tax revenue is the major source. Thus, in the analysis that follows we use the term taxes to indicate the total revenue generated by the government. As we all know, there is limited scope on the part of a government to increase tax revenue. The change in taxes depends upon several factors such as growth rate in gross domestic product (GDP), tax base, tax rate, etc.

Let us first consider a one-year decrease in taxes. Let us start from a situation where, until year 1, the government has balanced its budget, so that initial debt is equal to zero. During year 1, the government decreases taxes by 1 for one year. Thus, debt at the end of year 1, B_1 , is equal to 1. Let us consider various possibilities.

19.3.1 Full Repayment in Year 2

Suppose the government decides to fully repay the debt during year 2. From (19.2), the budget constraint for year 2 is given by

$$B_2 = (1 + r)B_1 + G_2 - T_2 \quad \dots(19.3)$$

If the debt is fully repaid during year 2, then the debt at the end of year 2 is equal to zero: $B_2 = 0$. Replacing B_1 by 1 and B_2 by 0 in equation (19.3) and transposing terms gives

$$T_2 - G_2 = (1 + r)1 = (1 + r) \quad \dots(19.4)$$

To repay the debt fully during year 2, the government must run a primary surplus equal to $(1 + r)$. It can be done in one of two ways, viz., (i) a decrease in spending, and (ii) an increase in tax revenue. We shall assume here and in the rest of this section that the adjustment comes through taxes, so that the path of spending is unaffected. It follows that the decrease in taxes by 1 during year 1 must be offset by an increase in taxes by $(1+r)$ during year 2.

19.3.2 Full Repayment in Year t

Now suppose the government decides to wait until year t to repay the debt. So, from year 2 to year $t-1$, the primary deficit is equal to zero—taxes are equal to spending. During year 2, the primary deficit is zero. So, from (19.3), debt at the end of year 2 is:

$$B_2 = (1 + r)B_1 + 0 = (1 + r)1 = (1 + r) \quad \dots(19.5)$$

Where the second equality uses the fact that $B_1 = 1$

With the primary deficit still equal to zero during year 3, debt at the end of year 3 is

$$B_3 = (1 + r)B_2 + 0 = (1 + r)(1 + r)1 = (1 + r)^2 \quad \dots(19.6)$$

Solving for debt at the end of year 4 and so on, it is clear that as long as the government keeps a primary deficit equal to zero, debt grows at a rate equal to the interest rate, and thus debt at the end of year $t-1$ is given by

$$B_{t-1} = (1 + r)^{t-2} \quad \dots (19.7)$$

Despite the fact that taxes are cut only in year 1, debt keeps increasing over time, at a rate equal to the interest rate. The reason is simple: Although the primary deficit is equal to zero, debt is now positive, and so are interest payments on the debt. Each year, the government must issue more debt to pay the interest on existing debt.

In year t , the year in which the government decides to repay the debt, the budget constraint is

$$B_t = (1 + r)B_{t-1} + (G_t - T_t) \quad \dots(19.8)$$

If debt is fully repaid during year t , then B_t (debt at the end of year t) is zero. Replacing B_t by zero and B_{t-1} by its expression from (19.7), in equation (19.8)

$$\text{gives } 0 = (1 + r)(1 + r)^{t-2} + (G_t - T_t) \quad \dots(19.9)$$

Reorganizing and bringing $(G_t - T_t)$ to the left side of the equation implies

$$T_t - G_t = (1 + r)^{t-1} \quad \dots(19.10)$$

To repay the debt, the government must run a primary surplus equal to $(1 + r)^{t-1}$ during year t . If the adjustment is done through taxes, the initial decrease in taxes of 1 during year 1 leads to an increase in taxes of $(1 + r)^{t-1}$ during year t .

This example yields our first set of conclusions:

- 1) If government spending is unchanged, a decrease in taxes must eventually be offset by an increase in taxes in the future.
- 2) The longer the government waits to increase taxes, or the higher the real interest rate is, the higher the eventual increase in taxes must be.

19.3.3 Debt Stabilization in Year t

We have assumed so far that the government fully repays the debt. Let us now look at what happens to taxes if the government only stabilizes the debt. (Stabilizing the debt means changing taxes or spending so that debt remains constant from then on). Suppose the government decides to stabilize the debt from year 2 on. Stabilizing the debt from year 2 on means the debt at the end of year 2 and thereafter remains at the same level as it was at the end of year 1. From equation (19.3), the budget constraint for year 2 is

$$B_2 = (1 + r)B_1 + G_2 - T_2$$

Under our assumption that debt is stabilized in year 2, $B_2 = B_1 = 1$. Setting $B_2 = B_1 = 1$ in the above equation yields

$$1 = (1 + r) + G_2 - T_2 \quad \dots(19.11)$$

Reorganizing, and bringing $(G_2 - T_2)$ to the left side of the equation,

$$T_2 - G_2 = (1 + r) - 1 = r \quad \dots(19.12)$$

To avoid a further increase in debt during year 1, the government must run a primary surplus equal to real interest payments on the existing debt. Taxes are permanently higher from year 1 on, by an amount equal to r : equivalently, from year 1 on, the government runs a primary surplus equal to r . Thus, the following conclusions follow:

- (i) To stabilize the debt, the government must eliminate the deficit.
- (ii) To eliminate the deficit, the government must run a primary surplus equal to the interest payments on the existing debt. This requires higher taxes forever.

19.3.4 Debt-GDP Ratio

We have focused so far on the evolution of the level of debt. But in an economy in which output grows over time, it makes more sense to focus instead on the *ratio of debt to GDP*. Let us derive the debt -GDP ratio first. Rewrite equation (19.2) as:

$$B_t = (1 + r)B_{t-1} + G_t - T_t \quad \dots(19.2b)$$

Now divide both sides of the above equation by real output, Y_t , to get

$$\frac{B_t}{Y_t} = (1 + r) \frac{B_{t-1}}{Y_t} + \frac{G_t - T_t}{Y_t} \quad \dots(19.13)$$

Next rewrite B_{t-1}/Y_t as $(B_{t-1}/Y_{t-1})(Y_{t-1}/Y_t)$. In other words, we multiply the numerator and the denominator by Y_{t-1} .

$$\frac{B_t}{Y_t} = (1 + r) \left(\frac{Y_{t-1}}{Y_t} \right) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad \dots(19.14)$$

Note that all the terms in the equation are now in terms of ratios to output, Y . To simplify this equation, we assume that output growth is constant and we denote the growth rate of output by g , so Y_{t-1}/Y_t can be written as $1/(1 + g)$. And use the approximation $(1 + r)/(1 + g) = 1 + r - g$. Using these two assumptions, let us rewrite the preceding equation (19.14) as

$$\frac{B_t}{Y_t} = (1 + r - g) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad \dots(19.15)$$

Finally, reorganize the above equation to get

$$\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}} = (r - g) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad \dots(19.16)$$

From equation (19.16) we find that the change in the debt ratio over time (the left side of the equation) is equal to the sum of two terms, viz., (i) The first term is the difference between the real interest rate and the growth rate times the initial debt ratio. The second term is the ratio of the primary deficit to GDP. Equation (19.16) implies that the increase in the ratio of debt to GDP will be larger:

- 1) The higher the real interest rate
- 2) The lower the growth rate of output, the higher the initial debt ratio
- 3) The higher the ratio of the primary deficit to GDP

Check Your Progress 2

- 1) What factors cause an increase in the debt to GDP ratio?

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- 2) What must the government do to stabilize the debt?

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19.4 THE DANGERS OF HIGH DEBT

Suppose that large deficits have led to a high debt ratio. What should the government do then? Simply trying to stabilize the debt at this high level is unwise: The lesson from history is that high debt can lead to vicious cycles and makes the conduct of fiscal policy extremely difficult. Let us look at this more closely.

Let us look at equation (19.16) again.

$$\frac{B_t}{Y_t} - \frac{B_{t-1}}{Y_{t-1}} = (r - g) \frac{B_{t-1}}{Y_{t-1}} + \frac{G_t - T_t}{Y_t} \quad \dots(19.16)$$

Let us take a country with a high debt ratio, say, 100%. Suppose the real interest rate is 3% and the growth rate is 2%. The first term on the right is (3%-2%) times 100% = 1% of GDP. Suppose further that the government is running a primary surplus of 1%, thus just enough to keep the debt ratio constant (the right side of the equation equals (3%-2%) times 100% + (-1%) = 0 per cent.

Now, suppose financial investors start to worry that the government may not be able to fully repay the debt. They ask for a higher interest rate, to compensate for what they perceive as a higher risk of default on the debt. But this in turn makes it more difficult for the government to stabilize the debt. Suppose, for example, that the interest rate increases from 3% to, say, 8%. Then, just to stabilize the debt, the government now needs to run a primary surplus of 6% (the right side of the equation is then equal to (8% - 2%) × 100% + (-6%) = 0 per cent.

Suppose that, in response to the increase in the interest rate, the government indeed takes measures to increase the primary surplus to 6 per cent. The spending cuts or tax increases that are needed are likely to prove politically costly, potentially generating more political uncertainty, a higher risk of default, and thus a further increase in the interest rate. Also, the sharp fiscal contraction is likely to lead to a recession, decreasing the growth rate. Both the increase in the real interest rate and the decrease in growth further increase (r - g), requiring an even larger surplus to stabilize the debt. At some point, the government may become unable to increase the primary surplus sufficiently, and the debt ratio starts increasing, leading financial markets to become even more worried and require an even higher interest rate. Increases in the interest rate and increases in the debt ratio feed on each other. The result is a debt explosion. The lesson is clear. When a government inherits a high debt ratio, it should aim at decreasing it over time. As (19.16) has made clear, it can achieve this through a combination of primary surpluses, high growth rates, and low real interest rates.

19.4.1 Debt Default

At some point, when a government finds itself facing very high interest rates, it may decide to default. Default is often partial, and creditors take what is known as a haircut: A haircut of 30 per cent, for example, means that creditors receive only 70 per cent of what they were owed. When debt is very high, default would seem like an appealing solution: Having a lower level of debt after default

reduces the size of the required fiscal consolidation and thus makes it more credible. It lowers required taxes, potentially allowing for higher growth. But default comes with very high costs. If debt is held, for example, by pension funds, the retirees may suffer very much from the default. If it is held by banks, then banks may go bankrupt, with major adverse effects on the economy. If debt is held instead mostly by foreigners, then the country's reputation may be lost, and it may be very difficult for the government to borrow abroad for a long time. So, in general, and rightly so, governments are very reluctant to default on their debt.

19.4.2 Lessons from the Past

High levels of government debt may increase the risk that an economy will experience capital flight that is an abrupt decline in the demand for a country's assets in world financial markets. International investors are aware that a government can always deal with its debt simply by defaulting. This approach was used as far back as 1335, when England's King Edward III defaulted on his debt to Italian bankers. More recently, several Latin American countries defaulted on their debts in the 1980s, and Russia did the same in 1998. The higher the level of the government debt, the greater is the temptation of default. Thus, as government debt increases, international investors may come to fear default and curtail their lending. If this loss of confidence occurs suddenly, the result could be the classic symptoms of 'capital flight' leading to currency crisis. It has occurred in Mexico in the early 1990s when default appeared likely. In 2011, it seemed likely that Greece was heading toward that outcome as well.

Greece's shaky financial status was well-known when it was admitted to the European Union (EU) in 2000, but the true depth of that nation's debt became known only in 2009 when Greece announced that it had been under-reporting its debt over the years. Meanwhile, fellow EU member Ireland, a strong financial performer, had to bail out its entire banking system in 2007-08. Between 2007 and 2010, the Irish debt-GDP ratio went up from 25 to about 100 percent, and its fiscal deficit as a percentage of GDP went up from zero per cent to more than 30 percent. At the same time, Portugal, Spain and Italy also started experiencing high debt problems. As the members of EU are connected, these nations' debt crisis threatened to trigger a sovereign debt crisis in Europe. Financial markets stopped viewing Italian, Greek, Irish, Portuguese and Spanish debt as close substitutes for German bonds, and they started hiking interest rates in the fall of 2008 to compensate for the heightened risk of default. By January 2012, the yield spread between German and Greek debt had increased by 33 percent. Greek banks then held about 20 percent of the government's sovereign debt. As the market feared Greece's banks would no longer honour obligations, they stopped rolling over Greek bank debt, which in turn meant that Greek banks could no longer roll over funding of the government's debt. A debt default seemed inescapable. To avert the crisis, the IMF, European Central Bank, and the European Commission agreed to extend emergency funding to Greece.

Since 2010, the EU and IMF have provided a total of more than €1 trillion in several loan packages to ease rollover problems, with Germany and France

contributing the most. The European Central Bank also injected up to €1 trillion of liquidity into the EU's banking system.

19.5 MONEY FINANCE

When the fiscal situation is bad, either because the deficits are large or the debt-GDP ratio is high, and the interest rate faced by the government is high, it becomes increasingly tempting for the government to want to finance itself through money finance. The government issues new bonds and instructs the central bank to buy them. The central bank then pays the government with the money it creates, and the government uses that money to finance its deficit. This process is called debt monetization.

How large a deficit can a government finance through such money creation? Let H be the amount of central bank money in the economy. Let ΔH be money creation; that is, the change in the nominal money stock from one month to the next. The revenue, in real terms that the government generates by creating an amount of money equal to ΔH is therefore $\Delta H/P$ – money creation during the period divided by the price level. This revenue from money creation is called seignorage.

$$\text{seignorage} = \frac{\Delta H}{P} \quad \dots(19.17)$$

To see what rate of (central bank) nominal money growth is required to generate a given amount of seignorage, we can rewrite $\Delta H/P$ as

$$\frac{\Delta H}{P} = \frac{\Delta H}{H} \frac{H}{P} \quad \dots(19.18)$$

From equation (19.8), we can think of seignorage ($\Delta H/P$) as the product of the rate of nominal money growth ($\Delta H/H$), and the real money stock (H/P). Thus we can re-write equation (19.17) as

$$\text{seignorage} = \frac{\Delta H}{H} \frac{H}{P} \quad \dots(19.19)$$

It is convenient to take one more step and divide both sides of the above equation by monthly GDP, say Y^I , to get:

$$\frac{\text{seignorage}}{Y^I} = \frac{\Delta H}{H} \left(\frac{H/P}{Y^I} \right) \quad \dots(19.20)$$

Suppose the government wants to finance a deficit of 10 per cent for a given ratio of central bank money to monthly GDP of 1. This implies that nominal money growth must satisfy

$$\frac{\Delta H}{H} \text{ times } 1 = 10\% \Rightarrow \frac{\Delta H}{H} = 10\%$$

From here we can infer that the monthly growth rate of nominal money of 10% is required to finance a deficit of 10 per cent. Unfortunately even such high rate of nominal money growth will not be sufficient to finance this deficit. As money growth increases, inflation is likely to follow. And very high inflation is likely to lead people to want to reduce their demand for money, and in turn the demand for central bank money. In other words, as the government increases ($\Delta H/H$), it is very likely that (H/P) will decrease. In response, the government needs to increase the rate of money growth further to achieve the same level of revenues. But higher money growth leads to further inflation, a further decrease in (H/P) and the need for further money growth. Soon, high inflation is likely to turn into

hyperinflation. You should note that several countries have witnessed hyperinflation in the past. Venezuela, for example, witnessed an inflation rate in the range of 3,00,000 per cent per year!

As inflation becomes very high, there is typically an increasing consensus that it should be stopped. Eventually, the government reduces the deficit and no longer has recourse to money finance. Inflation stops, but not before the economy has suffered substantial costs.

19.6 EFFECTS OF A DECREASE IN BUDGET DEFICIT

Suppose the government is running a budget deficit and decides to reduce it by decreasing its spending from G to G' while leaving taxes T unchanged. How will this affect the economy in the short run and in the medium run? Assume that output is initially at the natural level of output, so that the economy is at point A in Fig. 19.1. Output equals Y_n . The decrease in government spending from G to G' shifts the aggregate demand curve to the left, from AD to AD' . For a given price level, output is lower. In the short run, the equilibrium moves from A to A' ; output decreases from Y_n to Y' and the price level decreases from P to P' . The initial effect of the deficit reduction triggers lower output.

What happens over time? As long as output is below the natural level of output, we know that the aggregate supply curve keeps shifting down. The economy moves down along the aggregate demand curve AD' until the aggregate supply curve is given by AS'' and the economy reaches point A'' . By that time, the recession is over, and output is back at Y_n . At point A'' , not everything is the same as before: Output is back to the natural level of output, but the price level and the interest rate are lower than before the shift. The best way to see these specific effects is again to look at the adjustment in terms of the underlying $IS-LM$ model.

In panel (b) of Fig. 19.1, before the change in fiscal policy, the equilibrium is given by the intersection of the IS curve and the LM curve, at point A – which corresponds to point A in Fig. 19.1 panel (a). Output is equal to the natural level of output, Y_n , and the interest rate is given by i .

As the government reduces the budget deficit, the IS curve shifts to the left, to IS' . If the price level did not change, the economy would move from point A to point B . But, because the price level declines in response to the decrease in output, the real money stock increases, leading to a partially offsetting shift of the LM curve, down to LM' . So, the initial effect of deficit reduction is to move the economy from point A to point A' (Point A' in Fig. 19.1 panel (b) corresponds to point A' in Fig. 19.1 panel (a)).

Fig. 19.1 Panel (a): Decrease in Budget Deficit

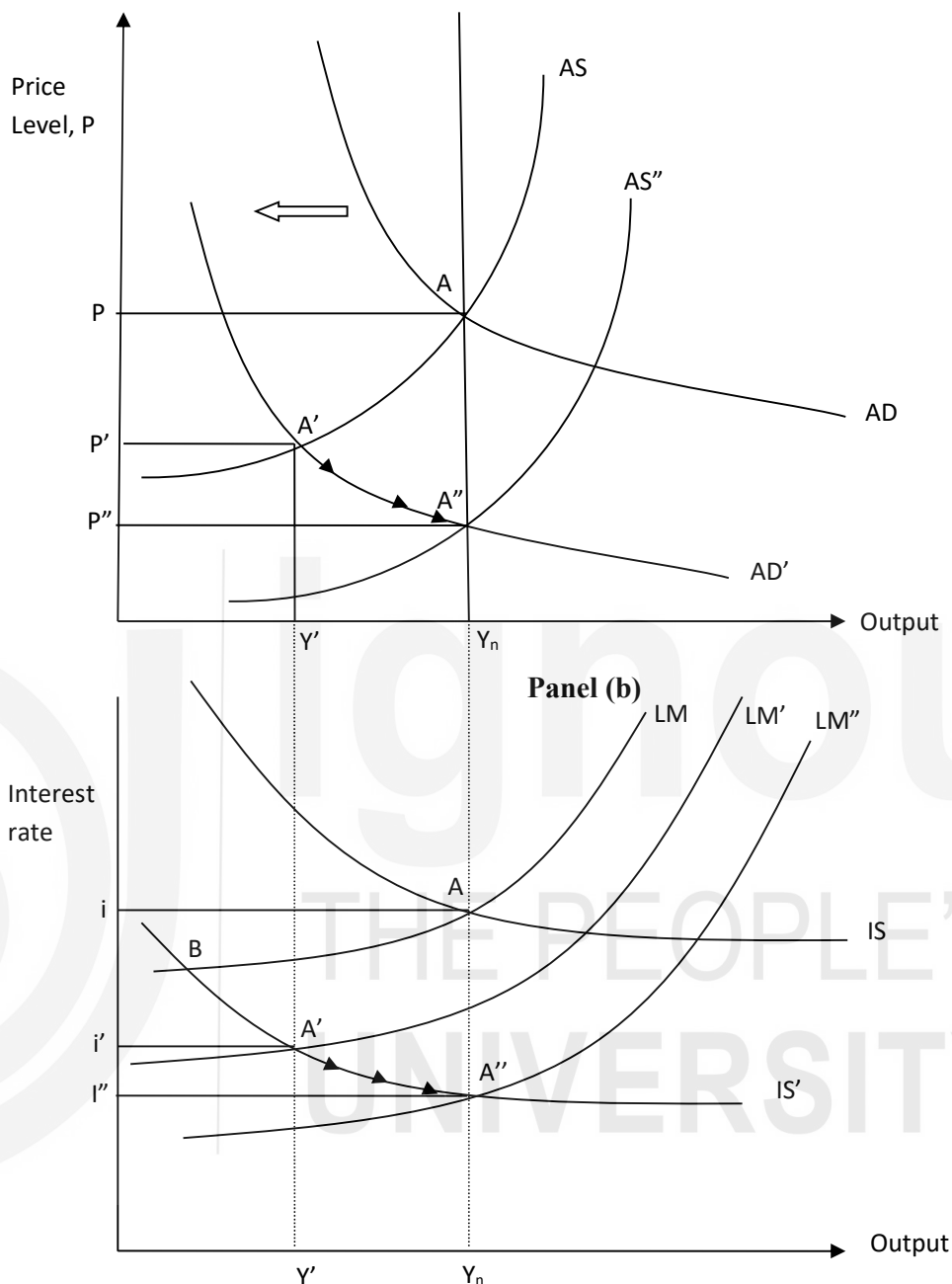


Fig. 19.1 shows that a deficit reduction leads in the short run to a decrease in output and to a decrease in the interest rate. In the medium run, output returns to its natural level, while the interest rate declines further.

Both output and the interest rate are lower than before the fiscal contraction. So long as output remains below the natural level of output, the price level continues to decline, leading to a further increase in the real money stock. The LM curve continues to shift down. In Fig. 19.1 panel (b), the economy moves down from point A' along IS' , and eventually reaches A'' (which corresponds to A'' in Fig. 19.1 panel (a)). At A'' , the LM curve is given by LM'' . At A'' , output is back at the

natural level of output. But the interest rate is lower than it was before deficit reduction, down from i to i'' . Hence, in the medium run, a reduction in the budget deficit unambiguously leads to a decrease in the interest rate and an increase in investment.

Check Your Progress 3

- 1) Will a high rate of nominal money growth be sufficient to finance the fiscal deficit through seignorage?

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19.7 LET US SUM UP

As the government carries out its fiscal policy through its budget, a closely related concept is of fiscal sustainability, which is loosely defined as the government's ability to meet its current and future obligations. It is the sustainability of government's finances. The issue of fiscal sustainability has taken on special importance in the aftermath of the global financial crisis. A sustainable policy is one where the ratio of debt held by the public to GDP (the debt-GDP ratio) is ultimately stable or declining. Fiscal sustainability can be achieved by putting consistent effort in reducing distortions and waste in a tax system and its administration, broadening the tax base, eliminating wasteful government expenditures, and providing public goods and services that achieve their goal in an efficient way.

19.10 ANSWERS/ HINTS TO CHECK YOUR PROGRESS

Check Your Progress 1

- 1) The government budget constraint links the change in government debt to the initial level of debt (which affects interest payments) and to current government spending and taxes.

Check Your Progress 2

- 1) The increase in the ratio of debt to GDP will be larger: The higher the real interest rate, the lower the growth rate of output, the higher the initial debt ratio, the higher the ratio of the primary deficit to GDP.
- 2) The government must eliminate the deficit by running a primary surplus equal to the interest payments on the existing debt.

Check Your Progress 3

- 1) As money growth increases, inflation is likely to follow. And very high inflation is likely to lead people to want to reduce their demand for money, and in turn the demand for central bank money. In other words, as the government increases, $\Delta H/H$, H/P is likely to decrease. As it does so, the government needs to increase the rate of money growth further to achieve the same level of revenues. But higher money growth leads to further inflation, a further decrease in H/P and the need for further money growth.

